

22.07.2024
LSM

$$Y = X\beta + \varepsilon \quad (X_i, Y_i) \quad i=1, 2, \dots, n$$

$\uparrow$
 $p \times 1$

Y_i scalar

$$\varepsilon = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}, Y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix} \quad \beta \in \mathbb{R}^p \quad X_{n \times p} = \begin{pmatrix} x_1^T \\ \vdots \\ x_n^T \end{pmatrix}$$

$$Y_i = x_i^T \beta + \varepsilon_i = \sum_{j=1}^p x_{ij} \beta_j + \varepsilon_i$$

$$= \sum_{j \neq j_0} x_{ij} \beta_j + \underline{x_{ij_0} \beta_{j_0}} + \varepsilon_i$$

β_{j_0} is the amount of change in Y_i for unit increase in the value of x_{ij_0}

Ex 1: $p=1 \quad x_i=1 \quad \mathbb{E}\varepsilon_i=0$

$$Y_i = \beta + \varepsilon_i$$

Ex 2 $Y_{11} \dots Y_{1n_1} \rightarrow$ Random (iid) sample from population 1 and 2
 $Y_{21} \dots Y_{2n_2}$ (P_1) (P_2)

Question: Are the two population the same?

$$P_1 \stackrel{?}{=} P_2$$

$$Y = X\beta + \varepsilon, \mathbb{E}\varepsilon = 0 \Rightarrow \mathbb{E}Y = X\beta$$

$$k=1,2 \quad Y_{ki} = \mu_k + \epsilon_{ki} \rightarrow \text{iid with mean 0}$$

$$Y = \begin{pmatrix} Y_{11} \\ \vdots \\ Y_{1n_1} \\ Y_{21} \\ \vdots \\ Y_{2n_2} \end{pmatrix}_{(n_1+n_2) \times 1} \quad i=1, \dots, n_k \quad k=1,2,\dots$$

$$\epsilon = \begin{pmatrix} \epsilon_{11} \\ \vdots \\ \epsilon_{1n_1} \\ \epsilon_{21} \\ \vdots \\ \epsilon_{2n_2} \end{pmatrix}$$

$$X\beta = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_1 \\ \mu_2 \\ \vdots \\ \mu_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ 0 & 1 \\ \vdots & \vdots \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \quad X\beta = \boxed{XA^{-1}} \boxed{A\beta}$$

rank 2

$$\mu_k = \theta + \delta_k$$

"pooled population mean"

$$\theta = w\mu_1 + (1-w)\mu_2$$

for some $0 < w < 1$

eg $w = \frac{n_1}{n_1+n_2}$

$$w\delta_1 + (1-w)\delta_2 = 0 \quad (\text{linear constraint})$$

$$Y = \begin{pmatrix} 1 & 1 & 0 \\ \vdots & \vdots & \vdots \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ \vdots & \vdots & \vdots \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \theta \\ \delta_1 \\ \delta_2 \end{pmatrix} + \epsilon$$

rank 2 (deficient) ("overparametrized", parameters are unidentifiable without further restriction)

Claim: In the description $Y = X\beta + \epsilon$ $\longrightarrow$ (*)
 $n \times 1 \quad n \times p \quad p \times 1 \quad n \times 1$

If $r = \text{rank}(X) < p$, then β (or the model given by *) is not identifiable and vice versa.

Not identifiable means, there exist $\beta^{(1)} \neq \beta^{(2)}$ s.t.

$$Y = X\beta^{(1)} + \epsilon = X\beta^{(2)} + \epsilon$$

$$X\beta^{(1)} = X\beta^{(2)} \iff X(\beta^{(1)} - \beta^{(2)}) = 0$$

$$\iff \beta^{(1)} = \beta^{(2)} \text{ unless } \text{rank}(X) < p$$

Gauss-Markov framework:

$$(X_i^{(1)}, \dots, X_i^{(p-1)}, Y_i) \quad i=1, \dots, n$$

$$Y_i = \beta_0 + \beta_1 X_i^{(1)} + \dots + \beta_{p-1} X_i^{(p-1)} + \epsilon_i$$

Intercept

β_j = slope for variable $X^{(j)}$ in the regression plane

Note: $\beta_0 + \sum_{j=1}^{p-1} \beta_j X^{(j)}$ is a linear function of

$(X^{(1)}, \dots, X^{(p-1)})$ it is also a linear function

of $\underline{\beta} = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_{p-1} \end{pmatrix}$

Ex. $X^{(j)} = X^j$, $j=1, \dots, p-1$ for some variable X
 Polynomial regression model

Estimating parameters in a linear model

Ex. One variable (simple) linear regression

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \quad (X_i, Y_i)$$

Estimate (β_0, β_1) by least ^{Scalar} squares procedure

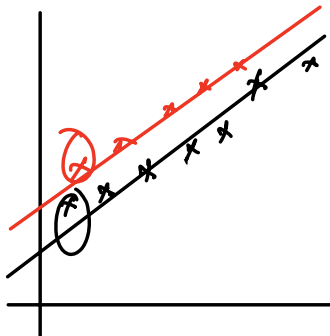
$$\min_{(\alpha_0, \alpha_1)} \sum (Y_i - \alpha_0 - \alpha_1 X_i)^2 \quad (\hat{\beta}_0, \hat{\beta}_1) \text{ are solution of this LS problem.}$$

$$e_i = Y_i - \hat{Y}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$$



$\varepsilon_i \sim \text{iid } N(0, 1)$
 passes through $(\bar{x}, \bar{y})$

* Randomness of order $n-1$



$$\varepsilon_i = 0.99\varepsilon_{i-1} + 0.01\delta_i$$

$$\delta_i \sim \text{iid } N(0, 1)$$

(AR1)

Randomness of order 1